

Spring Data JPA (Cognizant Digital Nurture) - Complete Project Implementation

Project Structure

```
src/main/java
├── com.cognizant.ormlearn
│   ├── OrmLearnApplication.java
│   ├── model/Country.java
│   ├── repository/CountryRepository.java
│   ├── service/CountryService.java
│   └── exception/CountryNotFoundException.java
src/main/resources
└── application.properties
```

pom.xml Dependencies

```
spring-boot-starter
spring-boot-starter-data-jpa
mysql-connector-j
spring-boot-devtools
```

application.properties

```
spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
spring.datasource.username=root
spring.datasource.password=root
spring.jpa.hibernate.ddl-auto=validate
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQL5Dialect
logging.level.org.hibernate.SQL=trace
logging.level.org.hibernate.type.descriptor.sql=trace
```

Country.java

```
@Entity
@Table(name="country")
public class Country{
    @Id
    @Column(name="co_code")
    private String code;
    @Column(name="co_name")
    private String name;
    // getters
    // setters
    // toString()
}
```

CountryRepository.java

```
@Repository
public interface CountryRepository extends JpaRepository<Country,String>{
    List<Country> findByNameContaining(String name);
}
```

CountryNotFoundException.java

```
public class CountryNotFoundException extends Exception{
    public CountryNotFoundException(String msg){
        super(msg);
    }
}
```

CountryService.java

```
@Service
public class CountryService{

    @Autowired
    private CountryRepository countryRepository;

    @Transactional
    public List<Country> getAllCountries(){
        return countryRepository.findAll();
    }

    @Transactional
    public Country findCountryByCode(String code)
    throws CountryNotFoundException{
        Optional<Country> result=countryRepository.findById(code);
        if(!result.isPresent())
            throw new CountryNotFoundException("Country Not Found");
        return result.get();
    }

    @Transactional
    public void addCountry(Country country){
        countryRepository.save(country);
    }

    @Transactional
    public void updateCountry(String code,String name)
    throws CountryNotFoundException{
        Country c=findCountryByCode(code);
        c.setName(name);
        countryRepository.save(c);
    }
}
```

```
}
```

```
@Transactional  
public void deleteCountry(String code){  
    countryRepository.deleteById(code);  
}
```

```
@Transactional  
public List<Country> searchCountry(String name){  
    return countryRepository.findByNameContaining(name);  
}  
}
```

OrmLearnApplication.java

```
@SpringBootApplication  
public class OrmLearnApplication{  
  
    private static final Logger LOGGER=  
        LoggerFactory.getLogger(OrmLearnApplication.class);  
  
    private static CountryService countryService;  
  
    public static void main(String[] args) throws Exception{  
  
        ApplicationContext context=  
            SpringApplication.run(OrmLearnApplication.class,args);  
  
        LOGGER.info("Inside main");  
  
        countryService=context.getBean(CountryService.class);  
  
        testGetAllCountries();  
        testFindCountry();  
        testAddCountry();  
        testUpdateCountry();  
        testSearchCountry();  
        testDeleteCountry();  
    }  
  
    static void testGetAllCountries(){  
        LOGGER.info("Start");  
        LOGGER.debug("{} ",countryService.getAllCountries());  
        LOGGER.info("End");  
    }  
  
    static void testFindCountry() throws Exception{
```

```
LOGGER.debug("{} ",countryService.findCountryByCode("IN"));  
}
```

```
static void testAddCountry(){  
    Country c=new Country();  
    c.setCode("ZZ");  
    c.setName("Demo Country");  
    countryService.addCountry(c);  
}
```

```
static void testUpdateCountry() throws Exception{  
    countryService.updateCountry("ZZ","Updated Country");  
}
```

```
static void testSearchCountry(){  
    LOGGER.debug("{} ",countryService.searchCountry("Ind"));  
}
```

```
static void testDeleteCountry(){  
    countryService.deleteCountry("ZZ");  
}  
}
```